keeping copies of many of its files, so they tasked me with engineering a solution and pitching it to the decision makers at headquarters. The result was a backup and storage system that would act as a shadow NSA: a complete, automated, and constantly updating copy of all of the agency's most important material, which would allow the agency to reboot and be up and running again, with all its archives intact, even if Fort Meade were reduced to smoldering rubble.

The major problem with creating a global disaster-recovery system—or really with creating any type of backup system that involves a truly staggering number of computers—is dealing with duplicated data. In plain terms, you have to handle situations in which, say, one thousand computers all have copies of the same single file: you have to make sure you're not backing up that same file one thousand times, because that would require one thousand times the amount of bandwidth and storage space. It was this wasteful duplication, in particular, that was preventing the agency's spoke sites from transmitting daily backups of their records to Fort Meade: the connection would be clogged with a thousand copies of the same file containing the same intercepted phone call, 999 of which the agency did not need.

The way to avoid this was "deduplication": a method to evaluate the uniqueness of data. The system that I designed would constantly scan the files at every facility at which the NSA stored records, testing each "block" of data down to the slightest fragment of a file to find out whether or not it was unique. Only if the agency lacked a copy of it back home would the data be automatically queued for transmission—reducing the volume that flowed over the agency's transpacific fiber-optic connection from a waterfall to a trickle.

The combination of deduplication and constant improvements in storage technology allowed the agency to store intelligence data for progressively longer periods of time. Just over the course of my career, the agency's goal went from being able to store intelligence for days, to weeks, to months, to five years or more after its collection. By the time of this book's publication, the agency might already be able to store it for decades. The NSA's conventional wisdom was that there was no point in collecting anything unless they could store it until it was useful, and there was no way to predict when exactly that would be. This rationalization was fuel for the agency's ultimate dream, which is permanency—to store all of the files it has ever collected or produced for perpetuity, and so create a perfect memory. The permanent record.

The NSA has a whole protocol you're supposed to follow when you give a